

Chapter 2	Configuration Space
Chapter 3	Rigid-Body Motions
	3.2 Rotations and Angular Velocities
	3.3 Rigid-Body Motions and Twists
Chapter 4	Forward Kinematics
Chapter 5	Velocity Kinematics and Statics
Chapter 6	Inverse Kinematics
Chapter 7	Kinematics of Closed Chains
Chapter 8	Dynamics of Open Chains
Chapter 9	Trajectory Generation
Chapter 10	Motion Planning
Chapter 11	Robot Control
Chapter 12	Grasping and Manipulation
Chapter 13	Wheeled Mobile Robots

Important concepts, symbols, and equations

- The **special Euclidean group** $SE(3)$ is a matrix Lie group also known as the group of rigid-body motions or **homogeneous transformation matrices** in $\mathbb{R}^3$.

$$T = \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_1 \\ r_{21} & r_{22} & r_{23} & p_2 \\ r_{31} & r_{32} & r_{33} & p_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \in SE(3) \quad R \in SO(3) \text{ and } p \in \mathbb{R}^3$$

- The inverse of $T \in SE(3)$ is

$$T^{-1} = \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} R^T & -R^T p \\ 0 & 1 \end{bmatrix}$$

Important concepts, symbols, and equations (cont.)

- Three uses of HT matrices:
 1. Represent a configuration. T_{ab} represents frame $\{b\}$ relative to $\{a\}$.
 2. Change the reference frame of a vector or frame.

$$T_{ab}T_{bc} = T_{a\cancel{b}}T_{\cancel{b}c} = T_{ac}$$
$$T_{ab}v_b = T_{a\cancel{b}}v_{\cancel{b}} = v_a$$

v should be written in
homogeneous coordinates,
 $v = [v_1 \ v_2 \ v_3 \ 1]^T$.

3. Displace a vector or frame. $T = (R, p) = \text{Trans}(p) \text{Rot}(\hat{\omega}, \theta)$

$$\text{Trans}(p) = \begin{bmatrix} 1 & 0 & 0 & p_x \\ 0 & 1 & 0 & p_y \\ 0 & 0 & 1 & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \text{Rot}(\hat{\omega}, \theta) = \begin{bmatrix} R & 0 \\ 0 & 1 \end{bmatrix}$$

Important concepts, symbols, and equations (cont.)

Space-frame transformation:

$$T_{sb}'' = T T_{sb} = \text{Trans}(p) \text{Rot}(\hat{\omega}, \theta) T_{sb}$$

1. rotate $\{b\}$ by θ about $\hat{\omega}$ in $\{s\}$ to get $\{b'\}$
(moves $\{b\}$ origin)
2. translate $\{b'\}$ by p in $\{s\}$ to get $\{b''\}$

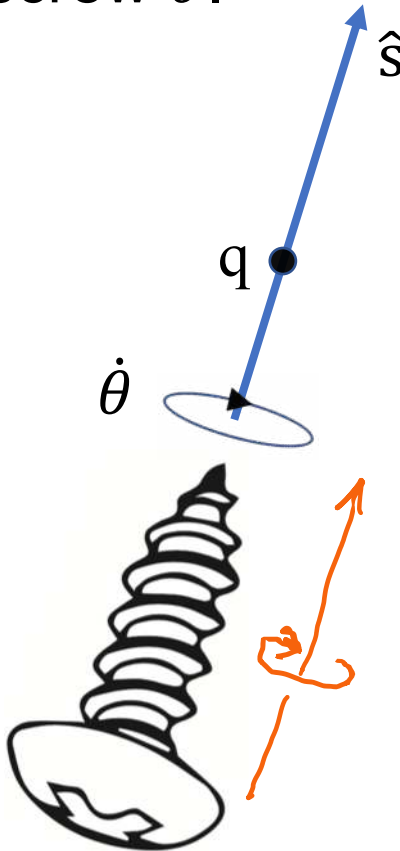
Body-frame transformation:

$$T_{sb}'' = T_{sb} T = T_{sb} \text{Trans}(p) \text{Rot}(\hat{\omega}, \theta)$$

1. translate $\{b\}$ by p in $\{b\}$ to get $\{b'\}$
2. rotate $\{b'\}$ by θ about $\hat{\omega}$ in $\{b'\}$ to get $\{b''\}$

Important concepts, symbols, and equations (cont.)

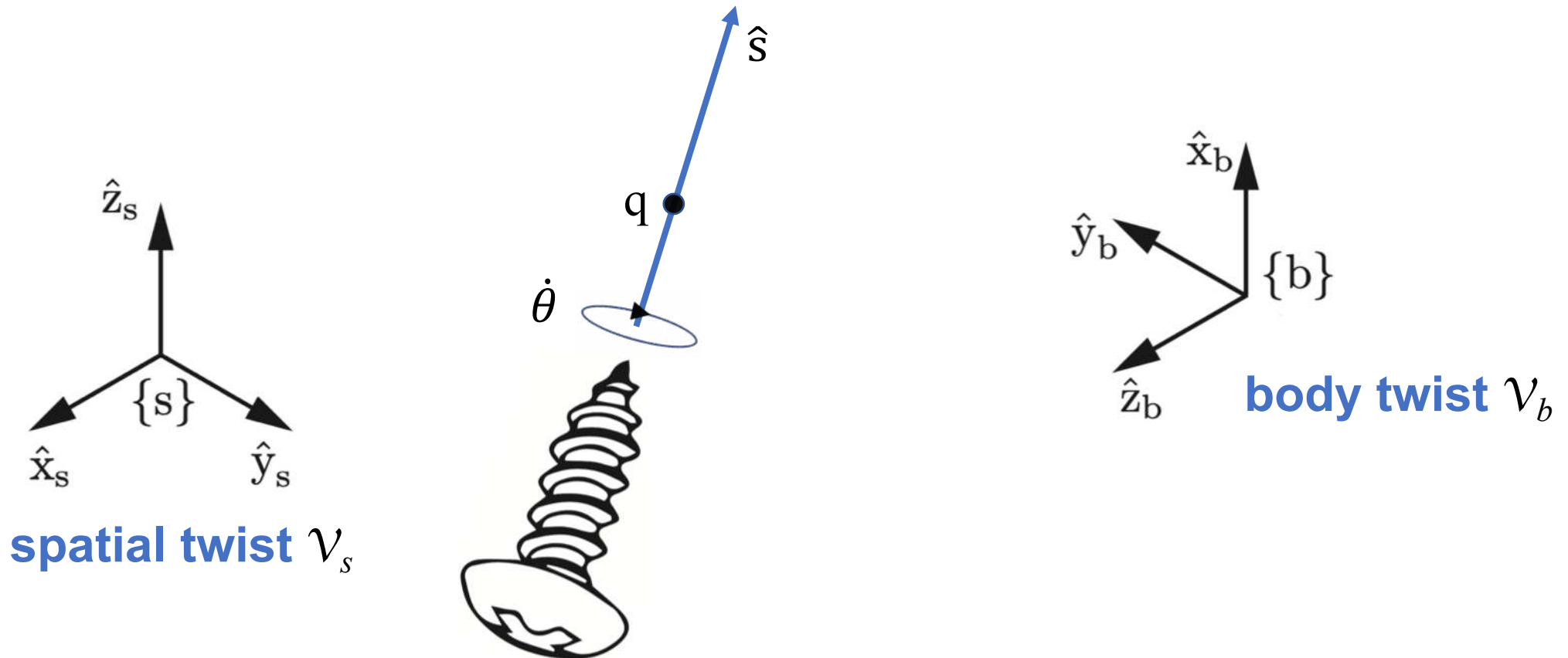
Any rigid-body velocity can be represented as a **screw axis** (a direction $\hat{s} \in S^2$, a point $q \in \mathbb{R}^3$ on the screw, and the **pitch** (linear speed/angular speed) of the screw h), plus the speed along the screw $\dot{\theta}$.



If h is infinite, $\dot{\theta}$ is the linear speed. Otherwise, it is the angular speed.

Important concepts, symbols, and equations (cont.)

The **twist** $\mathcal{V}_a = (\omega_a, v_a) \in \mathbb{R}^6$ is the angular velocity expressed in $\{a\}$ and the linear velocity of the origin of $\{a\}$ expressed in $\{a\}$.



Important concepts, symbols, and equations (cont.)

- To transform a twist from one frame to another,

$$\cancel{V_a = T_{ab} V_b}$$

$4 \times 4 \quad 6 \times 1$

$$V_a = [Ad_{T_{ab}}] V_b$$
$$\mathcal{J}_a = [Ad_{T_{ab}}] \mathcal{J}_b$$

$V_a = [Ad_{T_{ab}}] V_b$, where the **adjoint representation** of $T = (R, p)$ is

$$[Ad_T] = \begin{bmatrix} R & 0 \\ [p]R & R \end{bmatrix} \in \mathbb{R}^{6 \times 6}$$

Important concepts, symbols, and equations (cont.)

Matrix representation of twists:

4x4 matrix

$$T_{sb}^{-1} \dot{T}_{sb} = [\mathcal{V}_b] = \begin{bmatrix} [\omega_b] & v_b \\ 0 & 0 \end{bmatrix} \in se(3) \quad (\cancel{T_{bs}} \dot{\cancel{T_{sb}}})$$

$$\dot{T}_{sb} T_{sb}^{-1} = [\mathcal{V}_s] = \begin{bmatrix} [\omega_s] & v_s \\ 0 & 0 \end{bmatrix} \in se(3) \quad (\dot{\cancel{T_{sb}}} \cancel{T_{bs}})$$

where $se(3)$ is the Lie algebra of $SE(3)$ (the set of all possible $\dot{T}$ when $T = I$).

Important concepts, symbols, and equations (cont.)

Screws and twists:

- for a screw axis $\{q, \hat{s}, h\}$ with finite h ,

$$\mathcal{S} = \begin{bmatrix} \omega \\ v \end{bmatrix} = \begin{bmatrix} \hat{s} \\ -\hat{s} \times q + h\hat{s} \end{bmatrix}$$

- “unit” screw axis is $\mathcal{S} = \begin{bmatrix} \omega \\ v \end{bmatrix} \in \mathbb{R}^6$,

where either (i) $\|\omega\| = 1$ or

(ii) $\omega = 0$ and $\|v\| = 1$

- $\mathcal{V} = \mathcal{S}\dot{\theta}$

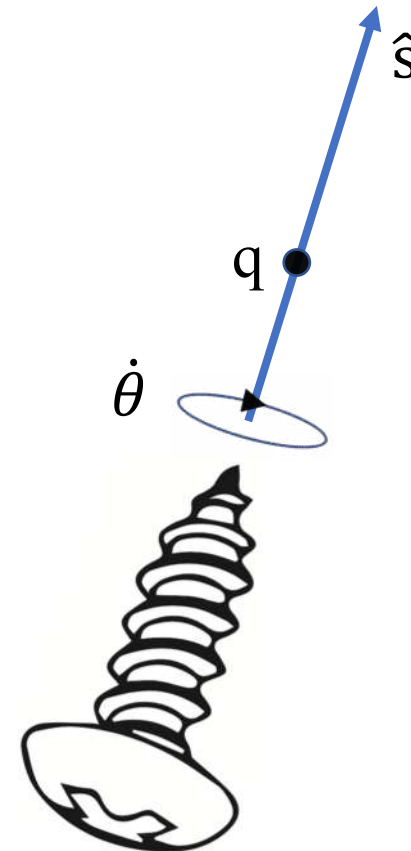
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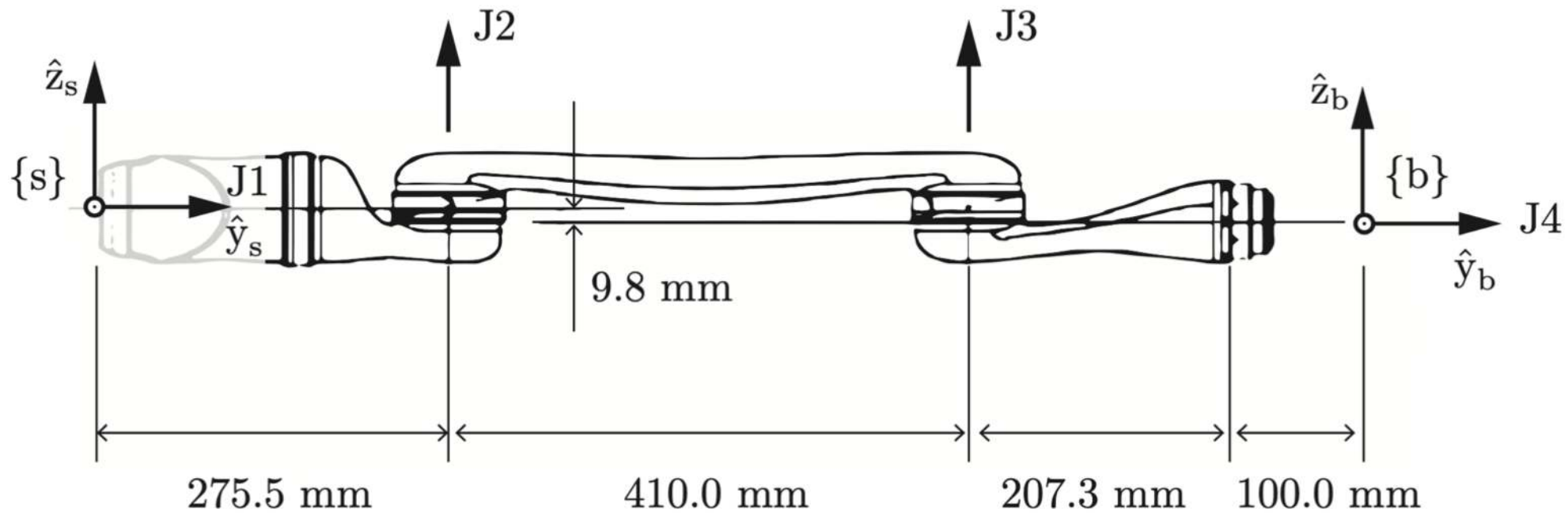
- $\mathcal{V} = \mathcal{S}\dot{\theta}$



What is the dimension of the space of screws?

5

Kinova lightweight 4-dof arm



For J4, what is the screw axis S_b ? S_s ?

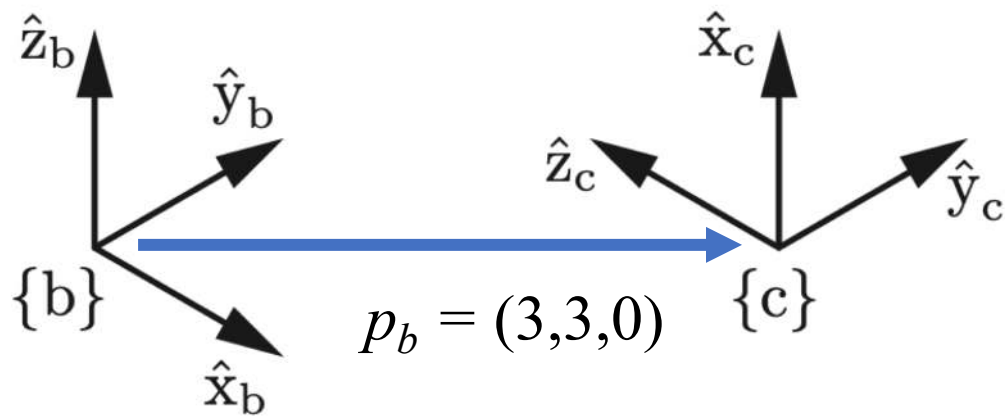
$$S_b = (\underbrace{0, 1, 0}_{\omega_b}, \underbrace{0, 0, 0}_{v_b})$$

$$S_s = (0, 1, 0, 9.8 \text{ mm/s}, 0, 0)$$

For J2?

$$S_b = (0, 0, 1, -717.3, 0, 0)$$

$$S_s = (0, 0, 1, 275.5, 0, 0)$$



Write T_{bc} .

$$T_{bc} = \begin{bmatrix} 0 & 0 & -1 & 3 \\ 0 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

A camera with frame $\{c\}$ tracks the optical marker on a tool to get its frame $\{t\}$ relative to $\{c\}$. This transformation is T_1 . A space frame $\{s\}$ is attached to the floor of the room, and the camera observes its configuration relative to $\{c\}$ as T_2 . A robot arm has a mounting frame $\{m\}$ which has been measured relative to $\{s\}$ as T_3 . The arm's encoders and the robot's kinematics tell us the gripper's frame $\{e\}$ relative to $\{m\}$. This is represented as T_4 . The gripper should be at the frame $\{g\}$ relative to $\{t\}$ to be able to close on the tool and pick it up. This configuration is represented as T_5 .

Write T_{eg} , the configuration of the grasping frame $\{g\}$ relative to the current end-effector frame $\{e\}$, in terms of T_1 , T_2 , T_3 , T_4 , and T_5 .

$$\begin{aligned}
 T_1 &= T_{ct} & T_3 &= T_{sm} & T_5 &= T_{tg} & T_{eg} &= T_{em} T_{ms} T_{sc} T_{ct} T_{tg} \\
 T_2 &= T_{cs} & T_4 &= T_{me} & & & &= T_4^{-1} T_3^{-1} T_2^{-1} T_1 T_5
 \end{aligned}$$